The Urea Cycle Research Hub: Automated Literature Curation and Multi-Parameter Variant Analysis Across Eight Disease Genes

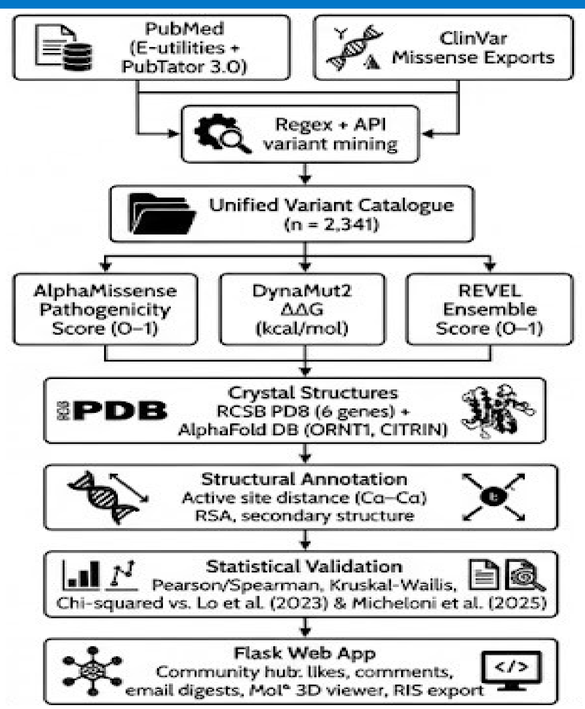
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INTRODUCTION

- To better characterize rare genetic disorders, a systematic method for collecting and integrating disparate information and most importantly, sharing this with the community is needed. We chose urea cycle disorders as an exemplar.
- Urea cycle disorders (UCDs) affect ~1 in 35,000 live births, causing hyperammonemia and neurological damage (Häberle et al., 2019)
- UCDs arise from pathogenic variants in 8 genes: OTC, CPS1, ASS, ASL, ARG1, NAGS, ORNT1, and CITRIN (Simpson et al., 2025)
- OTC deficiency is the most common UCD, with over 487 known missense variants: most uncharacterized experimentally (Caldovic et al., 2015)
- Computational tools like AlphaMissense (AM)(Cheng et al., 2023) and DynaMut2 (Rodrigues et al., 2021) offer pathogenicity and stability predictions, but require validation against lab tested, functional data
- **Goal: We developed the Urea Cycle Research Hub to be an accessible, community-based web platform integrating automated PubMed retrieval, ClinVar variant import, computational scoring, and 3D structural visualization to prioritize UCD variants for experimental validation. The platform can be readily modified to curate new diseases and easily extensible to include new data sources.**

METHODOLOGY



AM scores significantly correlated with experimental OTC functional results ($r = -0.446$, $p < 0.001$) and ClinVar classifications (Kruskal–Wallis $p = 1.58\times10^{-23}$)

59.5% of OTC variants had high AM scores but low thermodynamic destabilization: **OTC pathogenicity is most associated with changes to the active site**

Variants within 5 Å of the OTC active site **are the most correlated with OTC pathogenicity** (χ^2 $p = 1.24\times10^{-16}$)

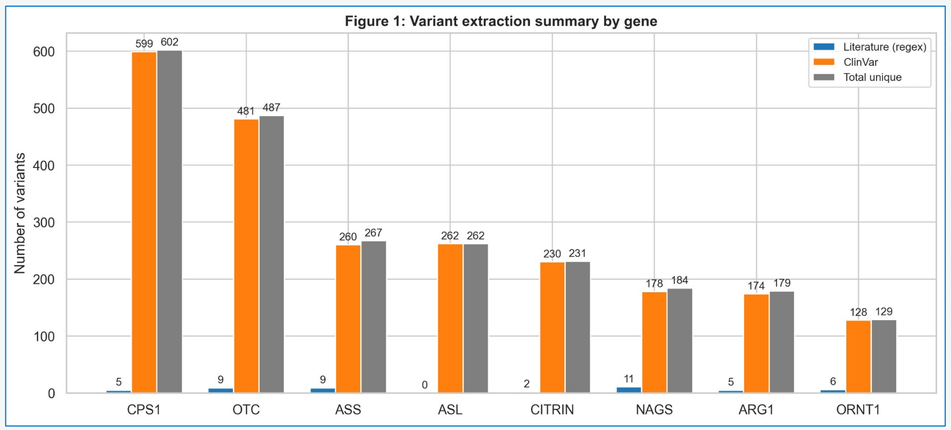


Figure 1. Variant extraction summary by gene. Bar chart showing literature-extracted (blue) and ClinVar-imported (orange) variant counts per gene. CPS1 yielded the largest set (n=602), ORNT1 the smallest (n=129). Total: 2,341 unique missense variants across all 8 genes.

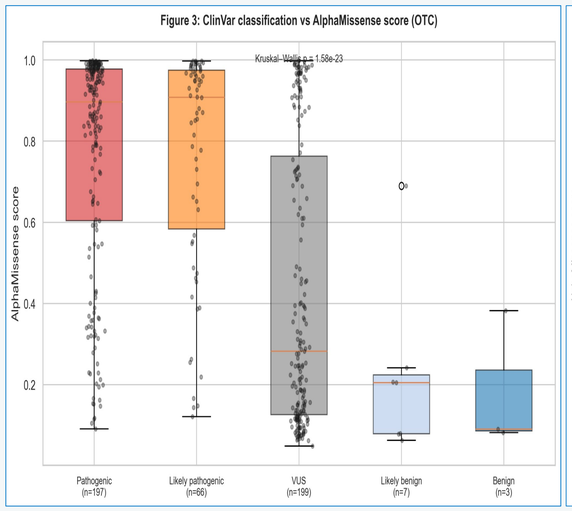


Figure 3. ClinVar classification vs. AlphaMissense score in OTC. Boxplots by clinical class (n=472 total). Pathogenic (median≈0.90) and likely pathogenic (median≈0.92) variants score higher than benign/likely benign (median≈0.20). VUS spans nearly the full 0–1 range. Kruskal-Wallis $p = 1.58\times10^{-23}$

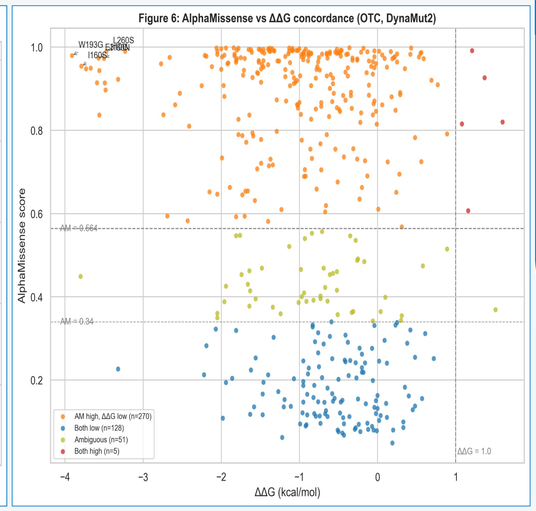


Figure 6. AlphaMissense vs. DynaMut2 ΔΔG concordance analysis for OTC. Points colored by category: AM high / ΔΔG low (orange, n=270, 59.5%); both low (blue, n=128, 28.2%); ambiguous (green, n=51); both high (red, n=5, 1.1%). Selected discordant variants labeled (W193G, L260S, E313N, I130N, I160S).

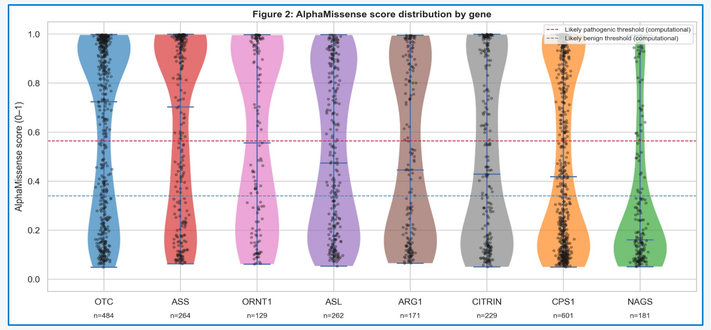


Figure 2. AlphaMissense score distributions across eight urea cycle genes. Violin plots with strip overlays by median score. OTC showed the highest median (≈0.73); NAGS the lowest (≈0.17). Dashed lines indicate the likely pathogenic (AM=0.564, red) and likely benign (AM=0.34, blue) thresholds (Cheng et al., 2023).

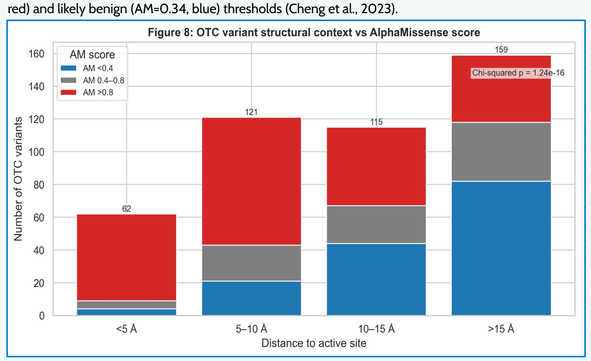


Figure 8. OTC variant AlphaMissense score distribution by distance to active site. Stacked bar chart binned by Cα–Cα distance to active site residues R92, R141, Q171, D263, S267, R330 (1C9Y). High-AM variants (AM>0.8, red) dominate the <5 Å bin; low-AM variants (AM<0.4, blue) increase proportionally with distance. Chi-squared $p = 1.24\times10^{-16}$.

CONCLUSION AND FUTURE DIRECTIONS

- AlphaMissense outperforms DynaMut2 ΔΔG for predicting OTC functional impact ($r = -0.446$ vs. 0.286), consistent with active site geometry disruption as the dominant pathogenic mechanism (Figures 6, 8)
- AlphaMissense performs best at functional extremes; moderately reduced variants (e.g., A152V, Y176C) are underestimated relative to kinetic assay data (Micheloni et al., 2025)
- VUS resolution requires combining AM, ΔΔG, REVEL, structural proximity, and ClinVar context; no single metric is enough for accurate prediction (Fig. 3)
- Limitations: Abstract-level variant extraction don't collect all variants in full-text mentions; ASL crystal structure flagged as lower-confidence (1AOS, 4.2 Å)
- Future directions:
  - Extend multi-parameter framework to all 8 UCD genes
  - Integrate LitVar 2.0 for full-text variant recovery (Allot et al., 2023)
  - Add RAG-based clinical decision support agent constrained to UCD literature corpus (Craig & Drăghici, 2024)

REFERENCES AND ACKNOWLEDGEMENTS

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